higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK

NQF LEVEL 4

(12030504)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

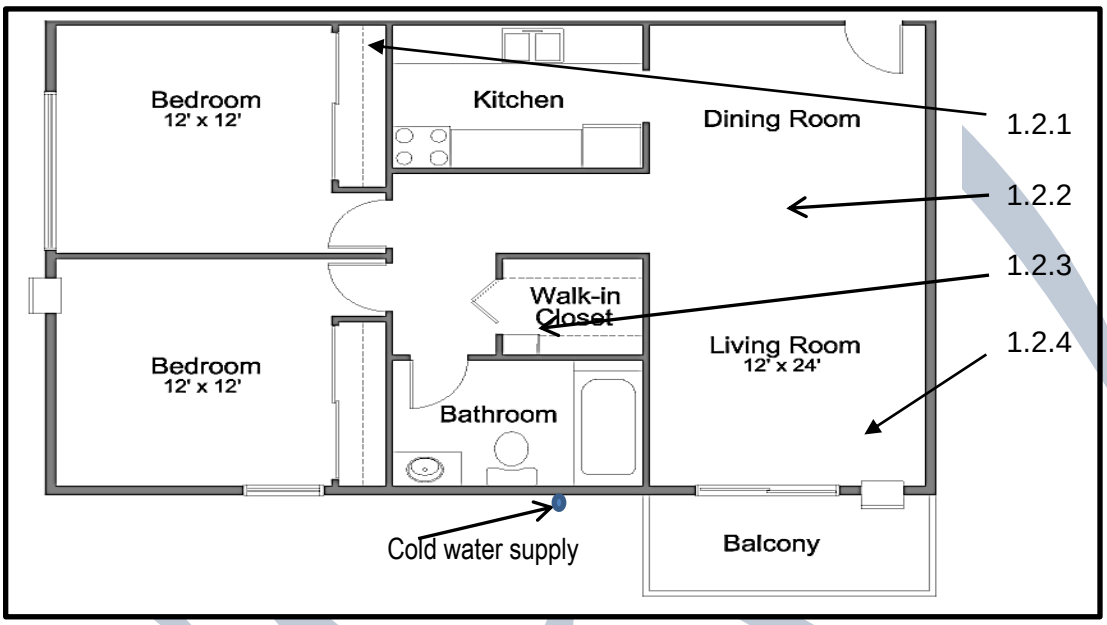
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QUESTION 1

1.1 How can trusses be strengthened to carry the load of a 250 ℓ geyser? (1)

1.2 FIGURE 1 below shows the plan view of a two bedroom house with a hipped roof structure. The plan is numbered on the right hand side (1.2.1-1.2.4) that indicate certain areas on the plan.

Give TWO reasons why you will or why you will not install the geyser support deck at each of the positions indicated as (1.2.1-1.2.4) in FIGURE 1. Write only the answer next to the question number (1.2.1–1.2.4) in the ANSWER BOOK.



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FIGURE 1

(4 x 2) (8)

1.3 How many trusses should be used to carry the load of a 250 ℓ geyser? (1)

1.4 A carpenter can build a geyser support deck on roof trusses without the supervision and approval of an engineer.

Explain when it is not permitted for a carpenter to design and install a geyser support deck. (2)

1.5 What are the advantages of placing a support deck on the internal walls rather than only on the bottom chords of the truss? (3)

[15]

QUESTION 2

2.1 Name FOUR functions of a ceiling in an office space. (4)

2.2 Give TWO advantages when installing gypsum ceiling boards. (2)

2.3 Explain the term *cornice* and the function of *cornices* when installing gypsum ceiling boards. (3)

2.4 Discuss the procedures that you will follow when installing a tongue and groove timber ceiling from the setting out of the ceiling up to the finishing. (20)

2.5 Explain why fibreglass insulation is best suited for ceilings? (1)
[30]

QUESTION 3

3.1 Give THREE advantages of a suspended timber floor over a concrete floor. (3)

3.2 A suspended timber floor is supported on the outsides by a one and a half brick foundation wall and a half brick sleeper wall on the inside. The suspended floor joists rest on sole plates with a damp-proof course under the sole plates. The foundation wall consists of 8 layers of bricks, supported by a foundation and the sleeper wall consists of 6 layers of bricks, supported by a concrete slab.

Make a neat labelled drawing, showing a vertical section of the suspended timber floor. The vertical section must only include the left side of the building up to the sleeper wall.

Include and label the following members in the drawing:

- Foundation
- Concrete slab
- Air brick
- Foundation wall
- Cavity wall
- Sleeper wall
- Damp proof course
- Ant guards
- Sole plates
- Floor joists
- Floor boards (20 x ½) (10)

3.3 Explain each of the following structural-engineering terms:

3.3.1 Mezzanine floor

3.3.2 Sole plate

3.3.3 Damp-proof membrane

(6 x ½) (3)

3.4 With the aid of a neat sketch, illustrate the secrete nailing method when fixing a tongue and groove floor board to a floor joist. (4)

[20]

QUESTION 4

4.1 Which TWO carpentry tools can be used to perform each of the following tasks during the installation of a flush door:

4.1.1 Trimming the door to the correct size and fitting it on the doorframe (2)

4.1.2 Chopping out waste material from the hinge-marked position (2)

4.1.3 Marking the position of the hinges on the door frame (2)

4.2 What is the function of the weather strip on the bottom rail of external doors? (2)

4.3 Write down a step-by-step description of how you would mark, trim and hang a hollow core door inside a wooden doorframe. (10)

4.4 Explain the term *latch bolt.* (4 x ½) (2)

[20]

QUESTION 5

5.1 Explain the term *moulded trim* when installing cupboards. (2)

5.2 Explain the use of a profile gauge. (2)

5.3 Name THREE materials commonly used for skirting boards. (3)

5.4 Explain the reason for the concave recess behind wooden skirting boards. (2)

5.5 Give THREE advantages and THREE disadvantages when using chipboard and melamine covered chipboard for the manufacturing and installation of cupboards. (6)

[15]

TOTAL: 100

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